

From Sample to Answer

The complete qPCR molecular workflow, the components each stage requires, and where the configuration choices actually matter. Written for laboratories bringing a method online, comparing suppliers, or working out what an existing workflow is missing.

5

Workflow stages,
end to end

3

Configurations
that change the work

1

Supplier across
every stage

ORIENTATION

Five stages, and the decisions inside them.

Every molecular detection workflow runs the same five stages. What separates a workflow that takes an afternoon from one that takes two days is not the science — it is how many discrete handling operations each stage costs, and whether the components were chosen to work together.

01

Specimen collection & transport

The matrix the assay will actually see is fixed here — and it constrains everything downstream.

02

Nucleic acid release

Purify, or release directly. The largest single fork in the workflow.

03

Reaction setup

Prepare and distribute, or receive reagents pre-dispensed. The second fork.

04

Amplification & detection

Channel capacity sets the ceiling on how many targets one reaction can resolve.

05

Controls & interpretation

Determines whether a negative result is a finding or an assumption.

Who this guide is for

Laboratory directors scoping a new method · QA and compliance staff auditing an existing workflow · research scientists specifying a panel · procurement consolidating a fragmented supply chain · anyone onboarding staff onto a molecular bench.

STAGE 01 OF 05

Specimen collection & transport.

The matrix is decided at this stage and cannot be renegotiated later. An assay validated in one matrix does not automatically carry to another, and transport chemistry can interfere with downstream steps.

WHAT THIS STAGE HAS TO ACHIEVE

- 1 Preserve the target**
 Nucleic acid has to survive collection, transit and storage in a state the assay can still detect.
- 2 Stay compatible downstream**
 Transport media, preservatives and stabilisers travel with the sample into extraction. Some inhibit.
- 3 Match the validated matrix**
 The matrix an assay was validated in is the matrix it can be relied on in. Substitutions need supporting data.

Where it goes wrong

-  A matrix in routine use that the assay was never validated against.
-  Transport medium introducing an inhibitor nobody accounted for.
-  Collection variability that only shows up as inconsistent Ct downstream.

BIOPATHOGENIX AT THIS STAGE



Collection supplies & transport media

Specimen collection devices and transport media, supplied alongside the assays that will consume them.



Biobank material

Over 1,000 heat-inactivated, authenticated, ATCC-traceable pathogens, available in blood, urine, Amies and Cary-Blair.



Consumables & PPE

Gloves, lab coats, masks, shoe covers, specimen bags and sharps containers on the same purchase order.

STAGE 02 OF 05

Nucleic acid release.

This is the stage most likely to be the bottleneck, and the one where an under-performing step produces the most dangerous result: a clean-looking negative. Some organisms actively resist release — thick-walled fungi, spores, and oocysts among them.

WHAT THIS STAGE HAS TO ACHIEVE

- 1 Break the organism open**
 Lysis has to be aggressive enough for the hardest target on the panel, not the easiest.
- 2 Remove or tolerate inhibitors**
 Whatever came in with the specimen is still there unless purification removed it.
- 3 Recover consistently**
 Run-to-run and operator-to-operator variation at this stage propagates into every downstream number.
- 4 Be demonstrable**
 An extraction control is what separates "no target present" from "extraction failed".

Where it goes wrong

-  A protocol tuned for easy targets, applied to organisms that resist lysis.
-  Manual handling variability at the highest-touch stage in the workflow.
-  A control that joins the workflow after extraction — and therefore cannot report on it.
-  Throughput that outgrows the method before anyone plans the transition.

BIOPATHOGENIX AT THIS STAGE



NucleoMini (BPX-16E)

Compact automated nucleic acid purification for smaller labs. Touchscreen control, small footprint, reduced plastic waste.



BPX Extraction Instrument

Automated 16-sample extraction with minimal consumable use, tested across bacterial, fungal and viral targets.



Extraction reagents

DNA and RNA extraction chemistry supplied and supported as part of the same workflow as the assays that follow it.

STAGE 02 · ALTERNATIVE

Or remove the stage entirely.

BPX™Direct-Detect is a two-reagent system that releases nucleic acid straight from the sample. There is no lysis-purify-recover sequence to run, no column or bead handling, and no purification instrument in the path.



DNA Direct-Detect

Reagents DNA-R1 / DNA-R2

For DNA-based organisms — bacteria, fungi, parasites and DNA viruses. Compatible with DNA polymerase.



Duo Direct-Detect

Reagents DUO-R1 / DUO-R2

Widens coverage to viral and RNA targets. Compatible with both reverse transcriptase and DNA polymerase.

WHAT CHANGES IN PRACTICE

- The purification stage disappears** No columns, no beads, no elution. Reagent addition replaces the sequence.
- Instrument comes out of the path** No purification instrument to schedule, maintain, or queue behind.
- Consumable load drops** Fewer tips, tubes and plastics consumed per sample processed.
- Inhibitor tolerance still matters** Because nothing was removed, the chemistry has to tolerate what remains. This is what the formulation is built for.

Both formulations ship as two reagents with a user manual, are lot-tested with a downloadable Certificate of Analysis, and are stored at 4 °C. Concordance and stability data are available on request.

STAGE 03 OF 05

Reaction setup.

Setup is where variability is introduced by hand. Every mix preparation, every distribution across a plate, and every pipetting decision is an opportunity for well-to-well difference that the data will later have to be explained around.

WHAT THIS STAGE HAS TO ACHIEVE

- 1 Deliver the right chemistry**
 Master mix chemistry has to suit the target type, the multiplex level, and whether reverse transcription is required.
- 2 Deliver it identically to every well**
 Uniform composition well to well is what makes replicates comparable and controls meaningful.
- 3 Fit the format the lab runs**
 Tube, 96-well or 384-well; full plates or partial runs; singleplex or multiplex.

Where it goes wrong

-  Mix prepared to different volumes by different operators on different days.
-  Distribution error across a plate that only surfaces as edge-well drift.
-  Reagent waste from preparing more mix than the run consumed.

TWO WAYS TO RUN THIS STAGE



Prepare it

Multiplex master mix and optimised primer and probe sets supplied as kit components, with a positive control and written protocols. Full control over reaction composition and volume.



Receive it pre-dispensed

Lyophilized qPCR plates arrive with reagents already in every well, in 96- and 384-well formats with a breakaway option for partial runs. Add sample and proceed to amplification.

Custom plate configuration is supported where no catalogue layout fits — target panel, multiplex configuration and plate map built to the experiment.

STAGE 04 OF 05

Amplification & detection.

Multiplex level is not an assay property alone — it is a property of the assay and the instrument together. A panel can only resolve as many targets in one reaction as the instrument has optical channels to read.

WHAT SETS THE CEILING

Q900 — eight optical channels

Eight channels is eight resolvable targets in a single reaction, from a single vial.

FAM

SIMA / HEX

TAMRA

CAL Fluor Red 610

CY5 / Quasar 670

CY5.5 / Quasar 705

CY7

CY7.5 / ICG

WHAT THIS STAGE REQUIRES

1

Enough channels for the multiplex level

A four-target panel needs four channels plus one for the internal control. Count the control.

2

A dye set the instrument can actually read

BPX qPLEX™ panels are supplied in channel configurations matched to common platforms rather than one fixed assignment.

3

A cycling program matched to the chemistry

Supplied with the kit protocol rather than left to be reconstructed locally.

Running on an existing instrument?

Send your platform and channel configuration to an assay specialist and the panel will be scoped against it before you order — support@biopathogenix.com

STAGE 05 OF 05

Controls & interpretation.

A negative result carries information only when the laboratory can demonstrate that extraction, amplification, reagents and instrumentation all behaved as expected. Controls are what convert a negative from an assumption into a finding.

THREE QUESTIONS A CONTROL SET HAS TO ANSWER

- 1 **Did the workflow run?**
Answered only by a control that entered before extraction and travelled the whole path.
- 2 **Did this particular sample run?**
Answered by an internal control carried inside the reaction, reporting per sample.
- 3 **Is the instrument reporting correctly?**
Answered underneath the workflow, by verifying cycling and optical uniformity across the block.

BIOPATHOGENIX AT THIS STAGE



qCHECK™ controls

External positive controls built from whole, intact, ATCC-traceable organisms that run through extraction, not just amplification. Lyophilized, rehydrated into your matrix.



RNase P Verification Plate

Instrument and channel performance check using the human RNase P target. Confirms thermal cycling, optical accuracy and well-to-well reproducibility. 96- and 384-well.



Contrived & split-sample

Characterised material built to specified concentrations and matrices, and parallel method comparison against an existing workflow.

CAP, COLA and CLIA expectations centre on a control that runs the entire testing process, extraction included.

CONFIGURATION

Three routes compared.

The same five stages, run three ways. The routes are not exclusive — Direct-Detect feeding a lyophilized plate removes handling from both forks at once.

	Standard extraction	Direct-Detect	Lyophilized plates
Purification stage	Required	Removed	Required
Master mix preparation	Required	Required	Not required
Reagent distribution	Required	Required	Not required
Cold-chain dependence	Per reagent spec	Refrigerated (4 °C)	Ambient
Instrument in the path	Purification system	None	Purification system
Best fit	Established workflows, mixed sample types	Hands-on time and throughput pressure	Cold storage limits, field and remote sites

Combining routes

Direct-Detect into a lyophilized plate removes the purification stage and the setup stage in the same workflow. Where no catalogue configuration fits, custom plate and panel development is supported — including target selection, multiplex configuration and plate layout.

DIAGNOSIS

Where workflows lose time.

Most of the delay in a molecular workflow is not in the chemistry. It accumulates in the seams — between suppliers, between stages, and between what a method was validated for and what it is now being asked to do.



Vendor coordination

Assays from one supplier, extraction from another, controls from a third, consumables from a fourth. Four lead times, four purchase orders, and no single party accountable when a result looks wrong.



The highest-touch stage left manual

Extraction is usually the queue. Where it stays manual while volume grows, it sets the ceiling on everything downstream.



Setup repeated by hand every run

Mix preparation and plate distribution repeated daily is variability introduced daily — and it lands in the data as well-to-well spread.



A control that cannot see the failure

An amplification-only control passing while extraction under-performed produces a negative nobody questions. This is the expensive one.



Validation treated as overhead

A method that cannot be validated within budget does not go live, and the capability sits unused. The constraint is commercial, not scientific.



Cold chain as a planning constraint

Dry ice, freezer allocation and shipment timing shape what a laboratory can run — particularly at remote, field and surveillance sites.

THE ECOSYSTEM

One supplier, every stage.

BioPathogenix manufactures the assays, holds the pathogen material, supplies the consumables and validates the methods. A workflow scoped here does not depend on a third party for any input it needs.

01

Collection

Collection supplies · transport media · PPE · specimen bags and sharps containers

02

Release

Extraction reagents · NucleoMini (BPX-16E) · BPX Extraction Instrument · DNA and Duo Direct-Detect

03

Setup

Multiplex master mixes · primer and probe sets · lyophilized 96- and 384-well plates · PCR consumables

04

Amplify & detect

Q900 multiplex qPCR system · BPX qPLEX™ panels in matched channel configurations

05

Control & verify

qCHECK™ external positive controls · RNase P Verification Plate · biobank · contrived samples

AND ACROSS EVERY STAGE

**Remote
Validation services**

All six regulatory study components

**Custom assay
development**

Built to your target list

**Direct technical
support**

Scientist to scientist

NEXT STEPS

Scoping a workflow.

A short scoping conversation is usually enough to map a target list, an instrument and a matrix onto a configuration. There is no cost attached to it.

WHAT TO HAVE READY

- The targets you need to detect, and whether the list is fixed or moving
- The sample matrices the assay will actually encounter
- Your real-time PCR platform and its channel configuration
- Current extraction method, and whether it is a throughput constraint
- Cold storage and cold-chain limitations, if any
- Your accrediting body — CLIA, CAP, COLA, or a combination

THREE WAYS TO START**Panel assessment**

Send your current target menu and instrument list. We map what a configuration would cover and where the gaps are, in writing.

**Validation consultation**

Scope a full method validation across all six regulatory study components, including how it would be funded.

**Custom panel scoping**

Where no catalogue configuration fits, we scope target selection, multiplex architecture and plate layout to the study.

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